RADL plan: guideline essentials

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What gets optimized

The project optimizes the convolution layers of a ResNet-8, its linear-algebra core, for one processing unit: an int8 conv accelerator in the FPGA fabric of the Avnet Ultra96 (Zynq UltraScale+), written in VHDL.

- **Offload:** the ARM A53 sends conv layers to the accelerator over AXI and keeps global pooling and the tiny 64→2 FC layer.
- **Dataflow:** output-stationary, output-parallel PE array of DSP MACs, with input line buffers in BRAM and all ≈ 77 k int8 weights resident in BRAM.

Optimizations:

1. v1 naive: 1 MAC unit, no buffering (hardware baseline).
2. BRAM buffering of weights and inputs, so fewer DRAM accesses.
3. Parallelism: unroll the 3×3 window (9 MACs), then 16 output channels (144 MACs).
4. Pipelining: one new MAC set per cycle.
5. Data types: int8 DSP packing; optionally lower bit-widths, with the accuracy effect measured.
6. Fusion: requant, ReLU and residual add in the write-back path.

Real-world application

Target: real-time cardiac view recognition (2-chamber vs 4-chamber) on a battery-powered wireless ultrasound probe. Siemens Healthineers ships real-time view recognition in its ACUSON Origin and keeps its ACUSON Freestyle wireless probes.

- **Why energy matters:** IEC 60601-2-37 caps the probe surface at 43 °C on a tissue-mimicking test object. AI compute shares that heat budget with the transducer, and in a wireless probe heat and battery both limit scan time.
- **Constraint:** ⩽ 33 ms per frame (30 fps, an assumption to verify) at batch size 1, since frames arrive one at a time.
- **Objective:** lowest energy per frame at that deadline.
- **Claims stay relative:** the Ultra96 stands in for the probe's embedded compute, so results read "X× less energy per frame than naive code on the same board, at the same accuracy".

Benchmark against a naive implementation

The naive baseline is plain C int8 inference (no libraries) on one ARM A53 core of the same board. It is compared with the optimized FPGA accelerator on the same int8 model and test set.

Metric	How it is measured
Latency per frame (ms), and whether ⩽ 33 ms holds	C timers on the A53; a cycle counter in the accelerator
Energy per frame (mJ)	DC power meter on the board supply
Test accuracy (%)	Same test set for both; target ⩾ 90 % after int8 (chance is 50 %)

Expected gap, ideal estimates from the layer shapes (37.4 M MACs per frame): naive C needs at least 31 ms even at 1 MAC per cycle, so it will likely miss the deadline. 64 parallel MACs at 200 MHz would take about 2.9 ms. Both get measured on the board.

Network, dataset and generality

The accelerator is tuned for one trained network but runs any conv layer of any network.

- **Network:** ResNet-8 with a stride-2 stem (16 → 32 → 64 channels, ≈ 37.4 M MACs, ≈ 77 k weights). Trained in Python, batch norm folded, quantized to int8, weights exported to C and VHDL.
- **Dataset:** CAMUS (open), 2,000 labeled frames, 2CH vs 4CH with 1,000 each, split by patient 400 / 50 / 50.
- **Not limited to this network:** layer sizes (C, K, H, W, kernel, stride, padding) are set at runtime through AXI-Lite registers, so the same accelerator and C framework handle any number of layers and any layer size.

Responsibilities

Each part has one owner, and that split is what the talk and report are organized around.

Owner	Work
	FPGA compute core: PE array, adder tree, accumulators; dataflow choice; int8 datapath and DSP packing; pipelining

Owner	Work
	System integration: AXI-Lite, DMA and Vivado block design; ARM driver and tiling; residual and pooling handling; benchmark and power harness
	Use case and dataset pipeline; training, quantization and export; naive C model; end-to-end evaluation

Open question: confirm each teammate's part once the group is fixed.

Final presentation and report

Each person presents and writes up their own part; the shared rows above go in the group part.

	Per participant (own work)	Group (overlapping parts)
Presentation	10 min	up to 10 min
Report	8–10 pages	up to 10 pages

Use the example slides in "Slides/Project Presentation" as the model for what the talk shows.